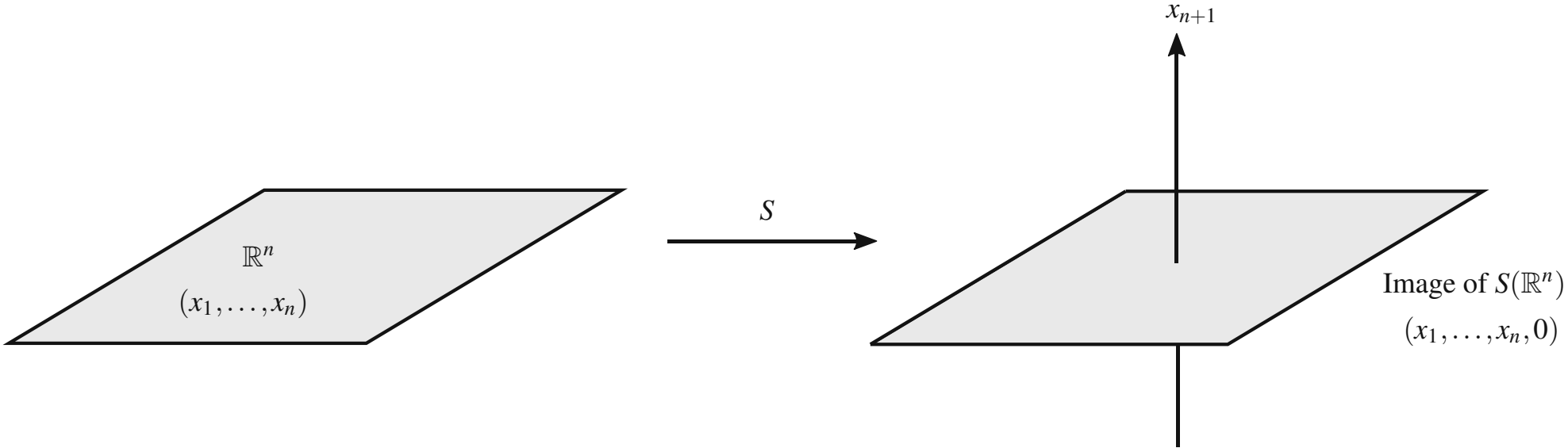

x_{n+1}
S
$\mathbb{R}^n$
$(x_1, \ldots, x_n)$
Image of $S(\mathbb{R}^n)$
$(x_1, \ldots, x_n, 0)$